

Soo Min Kwon

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EDUCATION

University of Michigan

Ph.D., Electrical Engineering and Computer Science

Sept. 2022 – May 2026

Ann Arbor, MI

- **Thesis:** “Deep Learning through Low-Dimensional Representations: Theory and Algorithms”
- **Advisor(s):** Prof. Laura Balzano and Prof. Qing Qu

Rutgers University

M.S., B.S., Electrical and Computer Engineering

Sept. 2016 – May 2022

New Brunswick, NJ

- **Thesis:** “Optimization Problems with Low-Dimensional Tensor Structures”
- **Advisor:** Prof. Anand D. Sarwate

EXPERIENCE

Google

Student Researcher

July 2025 – Nov. 2025

New York, NY

- Hosted by Himanshu Jain and Ziteng Sun
- Developed CoDistill-GRPO, a novel variant of GRPO aimed at improving small language model performance via co-distillation training with larger language models
- Tested the effectiveness of CoDistill-GRPO on the Qwen and Llama model families and with models up to 8B parameters

Amazon

Applied Scientist Intern

July 2024 – Nov. 2024

Seattle, WA

- Developed a causal inference framework using deep learning methods for the SCOT team that reduced variance estimates of average treatment effects by over 10%

LinkedIn

Applied Research Scientist Intern

May 2022 – Aug. 2022

Sunnyvale, CA

- Productionized a machine learning pipeline for the infrastructure team, reducing MAPE by over 15% in forecasting hardware needs for the next calendar year

SELECTED PUBLICATIONS

- [1] **Out-of-Distribution Generalization of In-Context Learning: A Low-Dimensional Subspace Perspective** [\[Paper\]](#)
S. M. Kwon[†], A. S. Xu[†], C. Yaras, L. Balzano, Q. Qu
AISTATS 2026; AISTATS 2026 CauScale Workshop (**Best Paper Award**); DeepMath 2025 (**Oral**)
- [2] **An Overview of Low-Rank Structures in the Training and Adaptation of Large Models** [\[Paper\]](#)
L. Balzano, T. Ding, B. D. Haeffele, **S. M. Kwon**, Q. Qu, P. Wang, Z. Wang, C. Yaras (α - β Order)
IEEE Signal Processing Magazine
- [3] **Learning Dynamics of Deep Matrix Factorization Beyond the Edge of Stability** [\[Paper\]](#)
A. Ghosh[†], **S. M. Kwon**[†], R. Wang, S. Ravishankar, Q. Qu
ICLR 2025
- [4] **BLAST: Block-Level Adaptive Structured Matrices for Efficient DNN Inference** [\[Paper\]](#)
C. Lee, **S. M. Kwon**, Q. Qu, H. Lee
NeurIPS 2024

- [5] **On the Relationship Between Small Initialization and Flatness in Deep Networks** [Paper]
S. M. Kwon, L. Ding, L. Balzano, Q. Qu
 ICLR 2024 Workshop on Bridging the Gap Between Practice and Theory in Deep Learning
- [6] **Efficient Compression of Overparameterized Deep Models through Low-Dimensional Learning Dynamics** [Paper]
S. M. Kwon, Z. Zhang, D. Song, L. Balzano, Q. Qu
 AISTATS 2024
- [7] **Solving Inverse Problems with Latent Diffusion Models via Hard Data Consistency** [Paper]
B. Song[†], S. M. Kwon[†], Z. Zhang, X. Hu, Q. Qu, L. Shen
 ICLR 2024 (**Spotlight, Top 5%**)

PREPRINTS

- [1] **CoDistill-GRPO: A Co-Distillation Recipe for Efficient Group Relative Policy Optimization** [Paper]
S. M. Kwon, Himanshu Jain, Ziteng Sun, Ananda Theertha Suresh, Sanjiv Kumar
- [2] **The Effect of Training Task Diversity on In-Context Learning through the Lens of Low-Dimensional Subspaces** [Paper]
S. M. Kwon, Alec S. Xu, Can Yaras, Dogyoon Song, Laura Balzano, Qing Qu
- [3] **Dynamic Subspace Estimation from Undersampled Data using Grassmannian Geodesics** [Paper]
S. M. Kwon[†], C. Blocker[†], H. Raja, J. Fessler, L. Balzano
- [4] **Decoupled Data Consistency for Solving General Inverse Problems with Diffusion Models** [Paper]
X. Li, S. M. Kwon, I. Alkhouri, S. Ravishankar, Q. Qu

AWARDS & HONORS

Harvey G. and Joyce H. Behner Graduate Fellowship	2024
University of Michigan PhD Rackham Merit Fellowship	2023
Rutgers ECE Outstanding Master's Student Award	2022
Rutgers ECE Outstanding Teaching Assistant Award	2021
Rutgers ECE Departmental Leadership & Service Award	2020
Rutgers WINLAB GA/TA Grant	2020
Rutgers University Dean's List	2018 – 2020

INVITED TALKS

2025 INFORMS Annual Meeting	2025
Math Machine Learning Seminar @ MPI MIS + UCLA	2025